

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424055
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SET 3: EMPLOYEE PERFORMANCE EVALUATION

Aim

To analyze employee performance using line, bar, and scatter charts, and to develop an interactive Tableau dashboard for exploring employee performance and department-wise distribution.

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1: Line Chart of Employee Performance

Question

Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

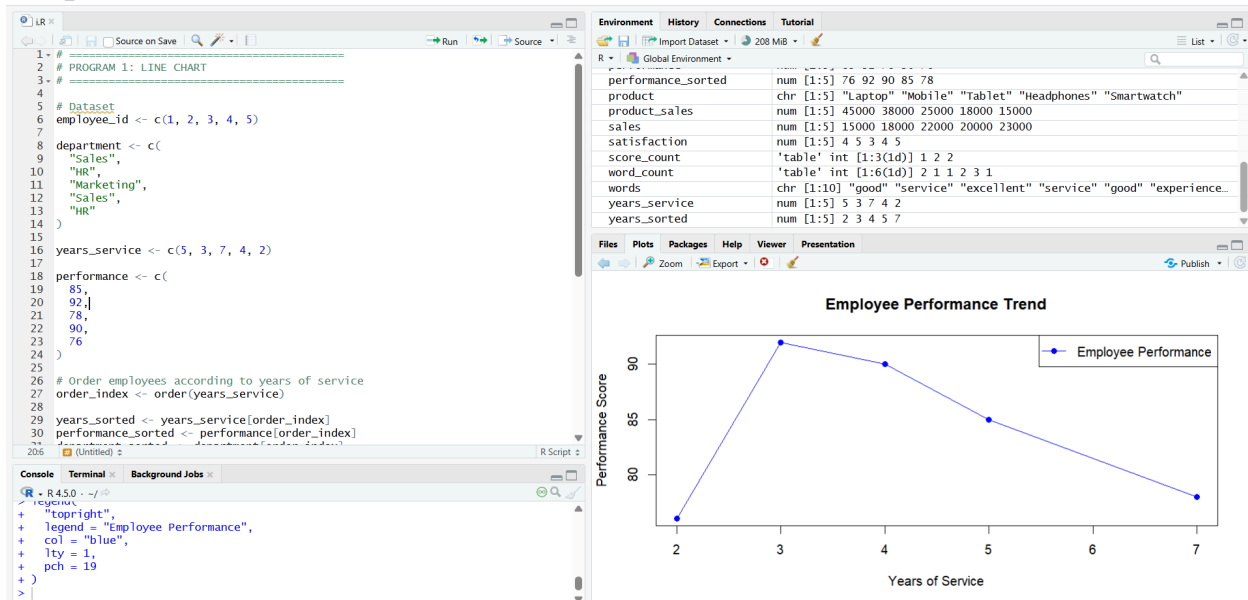
Since the dataset doesn't contain dates or years, **Years of Service** is used as the time/progression variable.

R Program

```
# =====  
# PROGRAM 1: LINE CHART  
# =====  
# Dataset  
employee_id <- c(1, 2, 3, 4, 5)  
department <- c(  
  "Sales",  
  "HR",  
  "Marketing",  
  "Sales",  
  "HR"  
)  
years_service <- c(5, 3, 7, 4, 2)  
performance <- c(  
  85,  
  92,  
  78,  
  90,  
  76  
)  
# Order employees according to years of service  
order_index <- order(years_service)  
years_sorted <- years_service[order_index]  
performance_sorted <- performance[order_index]  
department_sorted <- department[order_index]  
# Create line chart  
plot(  
  years_sorted,  
  performance_sorted,  
  type = "o",  
  pch = 19,  
  xlab = "Years of Service",  
  ylab = "Performance Score",  
  main = "Employee Performance Trend",  
  col = "blue"
```

```
)
# Add legend
legend(
  "topright",
  legend = "Employee Performance",
  col = "blue",
  lty = 1,
  pch = 19
)
```

Output



Observation

The performance scores vary across employees with different years of service. The highest performance score is 92 for the employee with 3 years of service, while the lowest score is 76 for the employee with 2 years of service.

PROGRAM 2: Bar Chart of Employees by Department

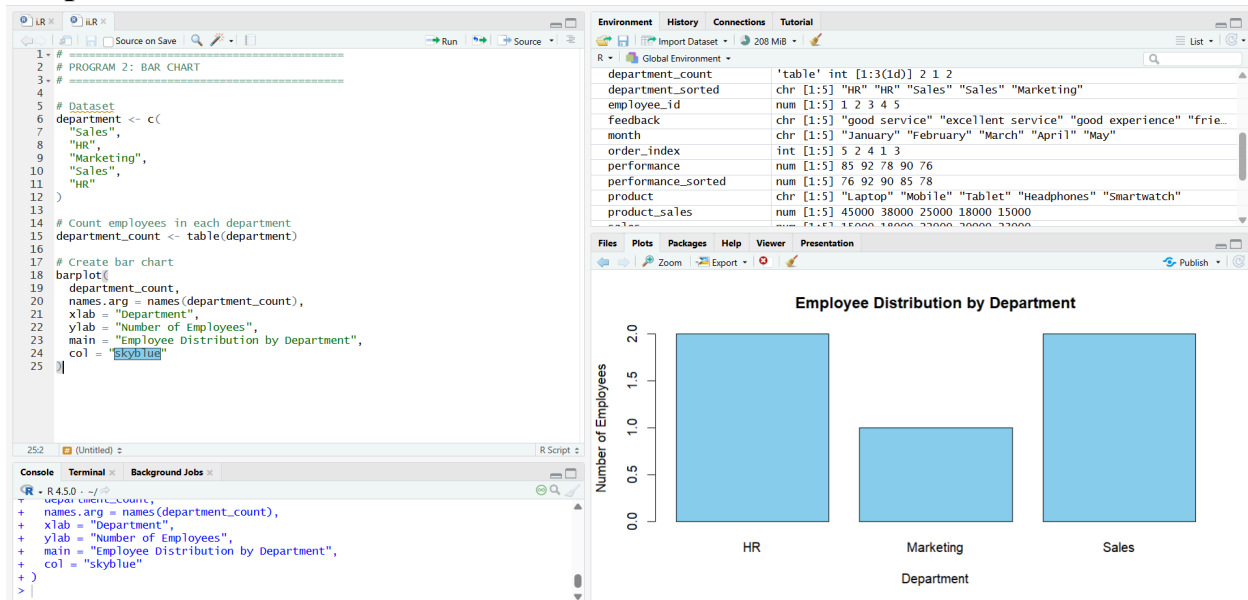
Question

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R Program

```
# =====  
# PROGRAM 2: BAR CHART  
# =====  
  
# Dataset  
department <- c(  
  "Sales",  
  "HR",  
  "Marketing",  
  "Sales",  
  "HR"  
)  
  
# Count employees in each department  
department_count <- table(department)  
  
# Create bar chart  
barplot(  
  department_count,  
  names.arg = names(department_count),  
  xlab = "Department",  
  ylab = "Number of Employees",  
  main = "Employee Distribution by Department",  
  col = "skyblue"  
)
```

Output



Observation

Sales and HR have the highest number of employees with two employees each, while Marketing has one employee.

PROGRAM 3: Scatter Plot — Years of Service vs Performance Score

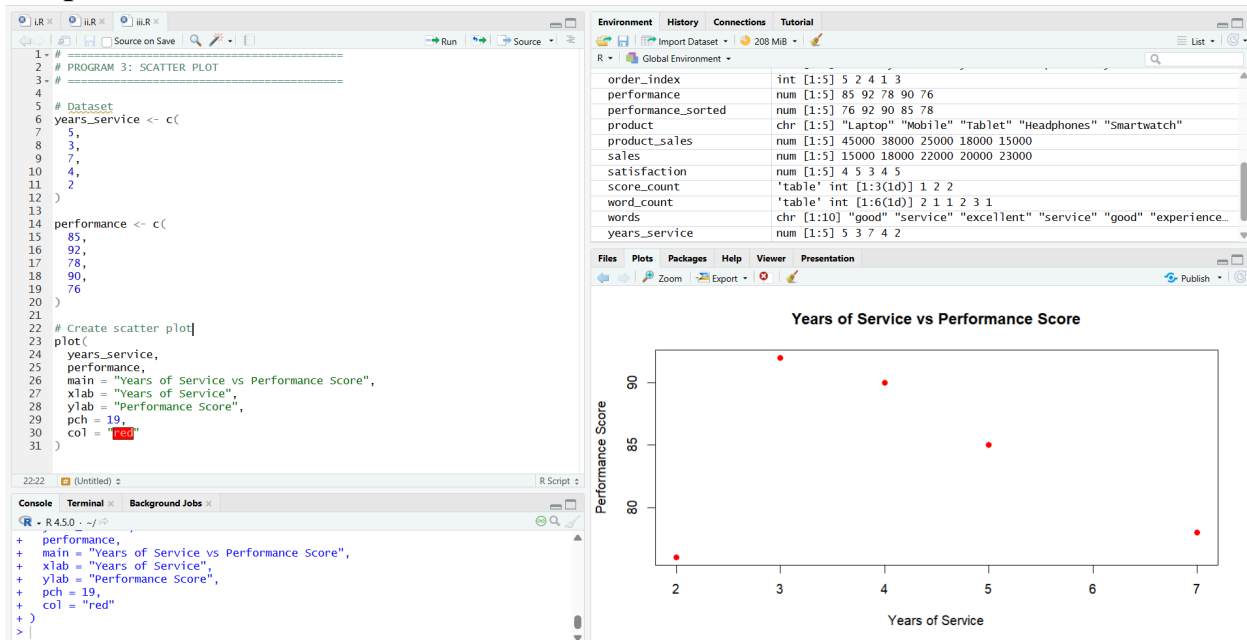
Question

Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

R Program

```
# =====  
# PROGRAM 3: SCATTER PLOT  
# =====  
  
# Dataset  
years_service <- c(  
  5,  
  3,  
  7,  
  4,  
  2  
)  
  
performance <- c(  
  85,  
  92,  
  78,  
  90,  
  76  
)  
  
# Create scatter plot  
plot(  
  years_service,  
  performance,  
  main = "Years of Service vs Performance Score",  
  xlab = "Years of Service",  
  ylab = "Performance Score",  
  pch = 19,  
  col = "red"  
)
```

Output



Insight

The scatter plot shows that performance scores vary across different years of service. In this small dataset, employees with more years of service do not consistently have higher performance scores. Therefore, years of service alone does not show a clear positive relationship with performance score in the given data.

PROGRAM 4: Tableau Dashboard

Question

Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Step 1 — Prepare your data

Create an Excel file with the following data:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Save it as:

Employee_Performance.xlsx

Step 2 — Open Tableau

Open **Tableau Desktop** or **Tableau Public**.

Choose:

Connect → Microsoft Excel

Then select:

Employee_Performance.xlsx

Step 3 — Create Line Chart

Go to **Sheet 1**.

Drag:

Years of Service



Columns

Drag:

Performance Score



Rows

Change the Marks type to:

Line

Drag:

Department

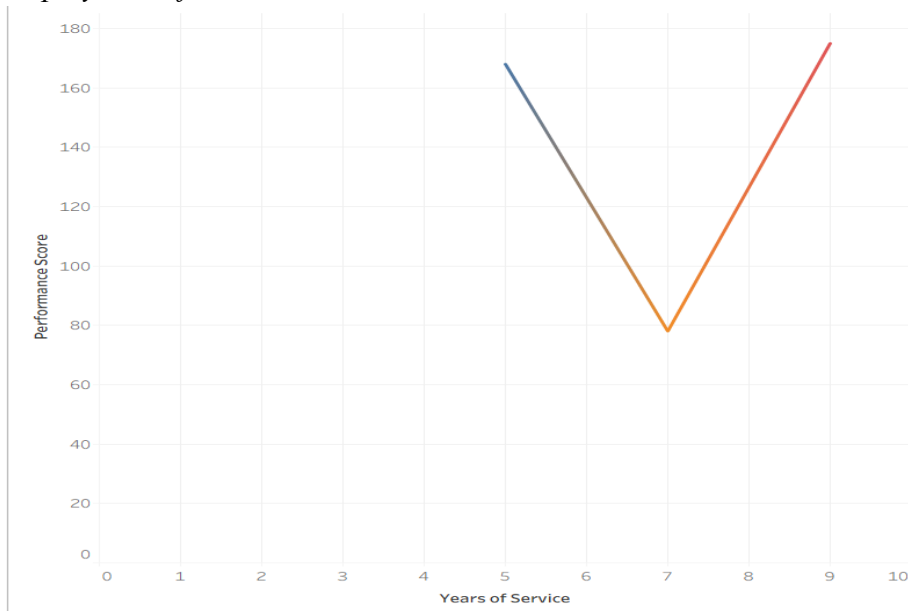


Color

This gives you a line chart with department information represented by the legend.

Chart title

Employee Performance Trend



Step 4 — Create Bar Chart

Create another worksheet.

Drag:

Department



Columns

Then use:

Employee ID



Rows

Change the aggregation of Employee ID to:

COUNT

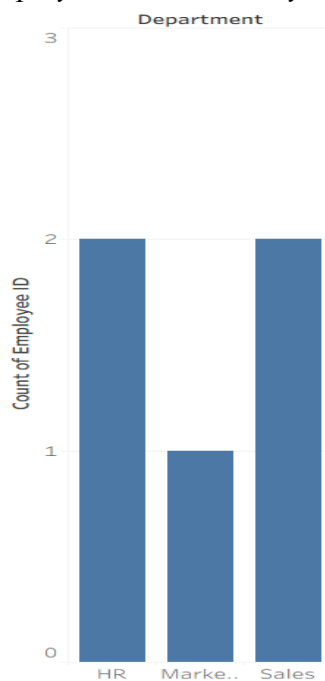
or use **Number of Records**.

Set Marks to:

Bar

Chart title

Employee Distribution by Department



The resulting chart should show:

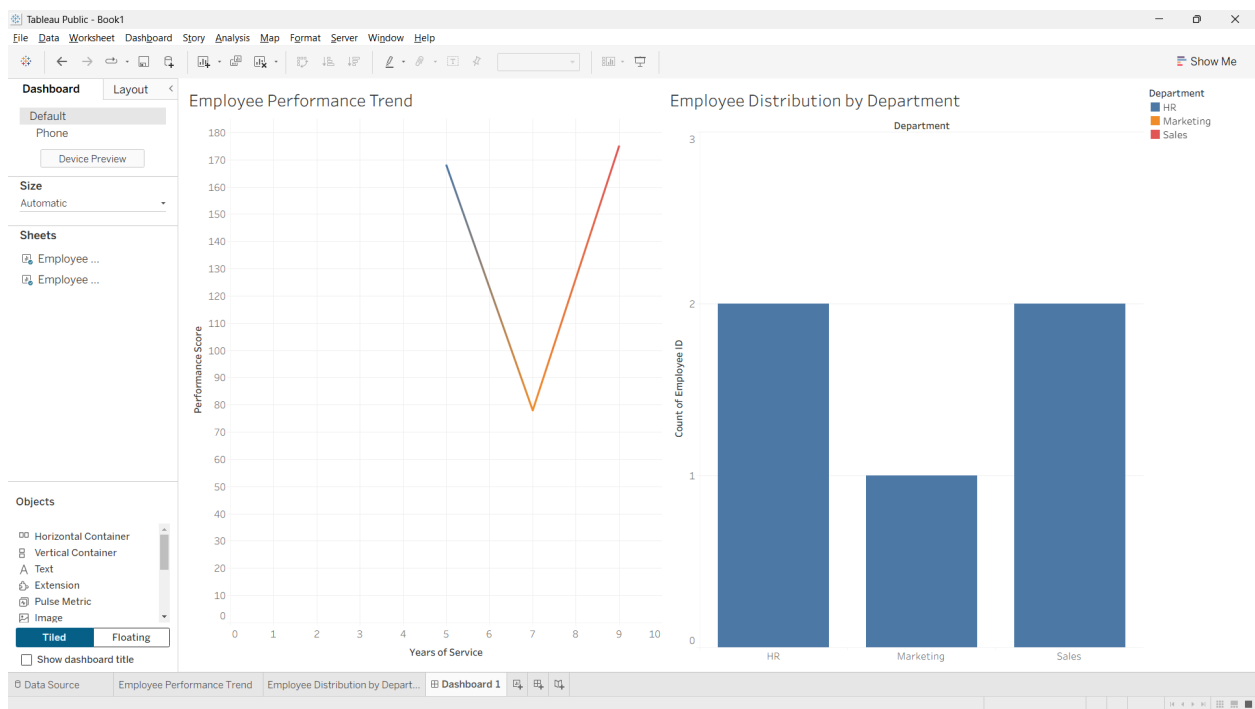
Sales 2
HR 2
Marketing 1

Step 5 — Create Dashboard

Click:

New Dashboard

Add both worksheets:



Observation

The Tableau dashboard combines the employee performance trend and department-wise employee distribution. Users can interact with the dashboard by selecting departments or applying available filters.

Procedure

1. Create the employee performance dataset.
2. Use Years of Service as the progression variable for the performance trend.
3. Create a line chart using Years of Service and Performance Score.
4. Add a legend to identify the employee performance series.
5. Count the employees in each department.
6. Use `barplot()` to create the department-wise employee distribution.
7. Use `plot()` to create the scatter plot between Years of Service and Performance Score.
8. Calculate the correlation between Years of Service and Performance Score.
9. Import the Employee Performance dataset into Tableau.
10. Create a line chart in Tableau.
11. Create a department-wise bar chart.
12. Combine both charts into an interactive Tableau dashboard.

Result

The employee performance dataset was successfully visualized using a line chart, department-wise bar chart, and scatter plot. The relationship between years of service and performance score was analyzed. An interactive Tableau dashboard was also created by combining the performance trend and department distribution charts.